

Battlefields of the Mind: rDCM rs-fMRI Insights into TBI and PTSD in Vietnam War Veterans

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1. Abstract

Traumatic Brain Injury (TBI) and Post-Traumatic Stress Disorder (PTSD) lead to structural and neurobiological changes potentially increasing the risk of late-life cognitive and affective decline, yet predicting individual trajectories remains challenging. In the Department of Defense Alzheimer's Disease Neuroimaging Initiative (DoD-ADNI), 88 Vietnam War veterans (aged 50–90; TBI = 26, PTSD = 20, TBI + PTSD = 9, NC = 33) underwent baseline 3 T structural MRI, resting-state fMRI, APOE genotyping, and neuropsychological testing, with follow-up after 12 months. Using the 246-region Brainnetome atlas, we extracted fMRI time-series to estimate undirected functional connectivity (FC) and directed effective connectivity via regression Dynamic Causal Modelling (rDCM). Unsupervised clustering (k-means, fuzzy C-means, Ward) consistently exceeded chance across dimensionality-reduction methods, indicating stable TBI-/PTSD-specific network disruptions. To predict 12-month Mini-Mental State Examination (MMSE), Clinical Dementia Rating–Sum of Boxes (CDR-SOB), and Geriatric Depression Scale (GDS-15) scores, we implemented a nested 5×3 -fold Mixture-of-Experts (MoE) framework combining questionnaire data, FC, rDCM, and APOE. Questionnaire-based models achieved test-set $R^2 = 0.295$ (MMSE), 0.640 (CDR-SOB), and 0.516 (GDS), while FC, rDCM, and APOE yielded non-positive R^2 over one year. These results demonstrate that atlas-driven connectivity robustly encodes NC, TBI, PTSD status but that near-term clinical outcomes are best forecast by baseline neuropsychological measures. Extending follow-up duration, increasing sample size, and integrating additional modalities may reveal the prognostic value of directed connectivity and genetic risk factors in aging veteran populations.

2. Background

Traumatic Brain Injury (TBI) and Post-Traumatic Stress Disorder (PTSD) have become increasingly pressing concerns because ongoing conflicts often expose military personnel and civilians to blasts, head injuries, and extreme psychological stress. TBI is an acquired brain insult resulting from an external mechanical force, such as a blow, jolt, or penetrating injury, to the head leading to a loss of consciousness sometimes with accompanying amnesia. PTSD is a psychiatric condition that can develop following exposure to a traumatic event involving actual or threatened death, serious injury, or sexual violence. Both conditions can lead to chronic physical and mental health difficulties, including persistent headaches, memory deficits, depression, anxiety, and personality changes, and many aspects of TBI pathology, especially structural white-matter and functional large-scale brain network disruptions, may contribute to future cognitive decline. In this context, the DOD ADNI project [9] enrolls Vietnam War veterans aged 50–90, with or without moderate-to-severe TBI or PTSD, alongside matched controls. By collecting longitudinal data over 12 months, including neuropsychological assessments, resting-state fMRI, APOE genotype and measures of everyday functioning, this study aims to determine whether specific connectivity patterns predict subsequent cognitive decline, depression or functional impairment. These findings could inform early detection strategies and targeted interventions for veterans at risk of neurodegenerative diseases.

2.1. Research Questions

- Are large-scale network disruptions identifiable using rs-fMRI?
- Can baseline rs-fMRI FC or rDCM predict 12-month cognitive/PTSD/functional decline?
- Do moderators like APOE or CSF biomarkers (amyloid, tau) influence these effects?

2.2. Clinical Outcome Measures

Neuropsychological assessments needed for baselines and follow-up: The Mini-Mental State Examination (MMSE) is a brief, 30-point questionnaire designed to screen for cognitive impairment by evaluating functions such as orientation, registration, attention, calculation, language, and visuospatial skills. In clinical settings, the MMSE score is interpreted according to in memory clinics widely used cutoffs: a score of 28 or higher for normal cognition; between 25 and 27 for mild cognitive impairment; 18 to 24 for with mild dementia; 10 to 17 moderate dementia; and scores of 9 or below indicate severe dementia. Thus, as the MMSE total declines from 30 toward zero, it reflects a progression from normal cognition through increasingly severe levels of impairment.

The Clinical Dementia Rating Scale – Sum of Boxes (CDR-SOB) provides a more granular staging of dementia by summing ratings (from 0 to 3) across six domains (memory; orientation; judgment and problem solving; community affairs; home and hobbies; and personal care) to yield a total score between 0.0 and 18.0. A CDR-SOB of 0.0 corresponds to no cognitive impairment, 0.5 to 4.0 reflects very mild impairment (equivalent to a global CDR of 0.5), 4.5 to 9.0 indicates mild dementia (global CDR of 1), 9.5 to 15.5 corresponds to moderate dementia (global CDR of 2), and 16.0 to 18.0 denotes severe dementia (global CDR of 3).

The Geriatric Depression Scale-15 (GDS-15) is a 15-item, self-report questionnaire used to screen for depressive symptoms in older adults. Each “yes/no” response indicative of depression scores one point, yielding a total between 0 and 15. According to the established cutoffs, a score of 0–5 points indicates no clinically significant depression, 6 to 10 points corresponds to mild-to-moderate depressive symptomatology, and 11–15 points reflects severe depressive symptomatology. This brief instrument typically takes about five minutes to complete and is widely validated for quickly identifying late-life depression.

3. Methods

3.1. Data selection

3.1.1. Dataset

In addition to the imaging data, a comprehensive set of study data is provided within the database, containing information deemed relevant and potentially useful for the research cohort. These supplementary data have been organized into several distinct directories, each representing a different domain of interest. Specifically, the available folders encompass diagnostic information, medical history, neuropsychological assessments, telephone-based pre-assessments and patient demographics, biospecimen results, and various imaging datasets (limited to MR and PET scans for a study subgroup). Additionally, logistical records, archived datasets, and administrative materials are included. Below, an overview is presented of the manner in which features for analysis were extracted.

3.1.2. Neuropsychological data

All files within the neuropsychological folder were systematically assessed to identify and extract relevant characteristics to be used in the generative embedding approach. Similarly, each file within the biospecimen folder was examined to determine the presence of disease defining cerebrospinal fluid measures and genetic risk factors that could potentially contribute to our study objectives. The diagnosis folder was used in conjunction with information from the enrollment, medical history, and subject characteristics folders, to clarifying each participant's diagnostic status, which enabled the categorization of subjects into four groups: PTSD, TBI, comorbidity

PTSD and TBI (PTSD+TBI), and matched normal controls (NC). It is important to note that the above files were reviewed in a thorough, manual process to assess the clinical condition of each subject and determine their final diagnosis and group assignment. The resulting diagnostic classifications and supporting comments can be found in the accompanying `sub-diagnosis.csv` file.

Within the neuropsychological folder, a detailed, file-by-file evaluation was undertaken. Initially, those files that consisted solely of responses to very specific questions in a qualitative manner without providing an overall or total score for the administered questionnaire were excluded, as limited utility for the intended analyses was offered by such files and the dataset would have been flooded with potentially irrelevant features. Files of a logistical nature, as well as one file identified to contain inconsistencies in participant responses or an excessive number of NaN values, were also excluded. For the remaining files, focus was placed on the total scores already calculated by the clinicians who conducted the assessments, with these summary measures being prioritized as most suitable for feature extraction. In the case of the Combat Exposure Scale (CES), however, all individual item columns were retained, their unique contribution to the study having been recognized.

Table 1 provides a comprehensive list of all features ultimately extracted from the neuropsychological folder, along with the exact variable names as they appear in the final CSV file.

For the one-year follow-up assessments, all available outcome measures offered by the study data for evaluating cognitive decline were initially considered: the MMSE (Mini-Mental State Examination) score, GDS (Geriatric Depression Scale), CDR-SOB (Clinical Dementia Rating Sum of Boxes), and FAQ (Functional Activities Questionnaire). However, the FAQ scores were excluded due to an excessive number of missing values. Consequently, the evaluation of cognitive decline and subsequent machine-learning prediction tasks were conducted primarily using the MMSE, because it is a widely employed measure of global cognitive function, with supplementary analyses based on GDS scores, due to validated sensitivity for late-life depressive symptoms, and CDR-SOB scores, because of detailed staging of the overall activities of daily living, obtained at the follow-up visit.

Imputation of missing neuropsychological data was performed using the mean value for each feature (i.e., each column) separately. Care was taken to avoid any implicit information leakage between the training and test sets, specifically, imputation was conducted independently within each partition, ensuring that the calculation of mean values for imputation in the test set was not influenced by the training data. The mean was chosen as a straightforward and transparent imputation strategy. Although multiple imputation by chained equations (MICE) was also considered as a more advanced alternative, this approach was ultimately not implemented due to limitations in MATLAB support and uncertainty regarding the availability and reliability of such functionality in Julia.

3.1.3. Imaging data

Participants who had a complete, high-resolution 3 Tesla T1-weighted (acquired using either an SPGR or MP-RAGE sequence) and a functional resting-state MRI scan at baseline were included. Scan parameters were highly consistent across subjects: the repetition time (TR) averaged 2.95 ± 0.045 s (range 2.90–3.00 s), the number of slices averaged 47.2 ± 2.4 (range 40–50), and the number of time points averaged 121.7 ± 20.2 (range 90–160). Voxel dimensions in the x and y axes both averaged 3.34 ± 0.16 mm (range 3.125–3.750 mm), while the z-axis averaged 3.38 ± 0.22 mm (range 3.30–4.00 mm).

3.1.4. Biospecimen

Within the biospecimen folder, genetic and molecular biomarker files were reviewed. It was determined that only the APOE genotype variables (APGEN1 and APGEN2) possessed adequate data coverage and demonstrated relevance based on the literature; therefore, these variables were retained as features. Clinical meaningful cere-

brospinal fluid measures ABETA, TAU, PTAU, had substantial proportions of missing data and were therefore not included. All other biospecimen variables were excluded due to excessive missing values. Consequently, only subjects with available APOE information were included, resulting in a slight reduction in the overall sample size. This also directly implies that no data imputation was needed.

3.1.5. Inclusion criteria

To determine the eligible subjects, participants with good-quality baseline structural and functional MRI imaging data were first selected. Afterwards, scores on key neuropsychological measures from baseline and 12 months follow-up visit, specifically the GDS, MMSE, and CDR-SOB scales, were examined. The intersection of subjects with both valid imaging and complete follow-up scores on these cognitive and functional assessments was taken to ensure that the final cohort included only individuals for whom cognitive decline could be meaningfully tracked over the one-year follow-up period.

Source	Used Features (column names)
Clinical Dementia Rating Scale, Sum of Boxes	CDR-SOB
Geriatric Depression Scale (GDS-15)	GDS15
Mini-Mental State Examination (MMSE)	MMSE
Clinician Administered PTSD Scale – Lifetime (CAPSLIFE)	CAPSSCORELIFE
Clinician Administered PTSD Scale – Current (CAPSCURR)	CAPSSCORECURR
Combat Exposure Scale (CES)	CES1A, CES2, CES3A, CES4, CES5, CES6, CES7
Neuropsychological Battery (NEUROBAT)	ANARTERR
sub_diagnosis_n88.csv	PATIENTGROUP
basic_imaging_information_for_every_subject.csv	Age, Weight

Table 1. Neuropsychological and biospecimen files and columns used for feature selection. All features are at screening or baseline.

3.2. Descriptive statistics

After applying the predefined inclusion criteria, a total of 88 participants were retained for analysis. Demographic (age) and clinical (CDR-SOB, GDS-15, MMSE, and CAPS) variables were summarized for each diagnostic group (NC, PTSD, TBI, PTSD+TBI). One-way ANOVAs were conducted to assess group differences across these measures.

3.3. rsfMRI Preprocessing and sanity checks

After manual inspection of the imaging data, volume-based resting-state preprocessing was performed using DPABI [17], which integrates the DPARSF pipeline [6] built on the SPM core routines [1]. After discarding the first ten volumes to allow magnetization to reach steady state, functional images underwent head-motion correction via a two-pass, six-parameter rigid-body realignment. Structural T1 images were then co-registered to the mean functional image and segmented into gray matter, white matter, and cerebrospinal fluid [3]. All images were normalized to MNI-152 template space using DARTEL [11] and Friston-24 [2] motion correction model was applied to regress out residual head-motion effects. Additional nuisance regressors included mean white-matter and CSF signals, as well as linear trends. Finally, voxel time-series were bandpass filtered (0.01–0.1 Hz) to suppress low-frequency drift and high-frequency physiological noise.

3.4. Brain Connectivity Estimation

Functional and effective connectivity estimation techniques were applied to time series extracted from the preprocessed baseline rs-fMRI data, with ROIs defined using the Brainnetome Atlas [14].

3.4.1. Functional Connectivity Estimation

Functional Connectivity (FC) is estimated using pairwise Pearson correlations between regional time series extracted from rs-fMRI to establish a benchmark for assessing network-level interactions (Fig. 2). A statistical view of temporal co-activation across brain regions is offered by this method without modeling the directionality of interactions [20]. Although directionality or causality is not inferred by FC, a baseline for comparison against rDCM is provided, and a control measure is served when the predictive power of directed connectivity is evaluated.

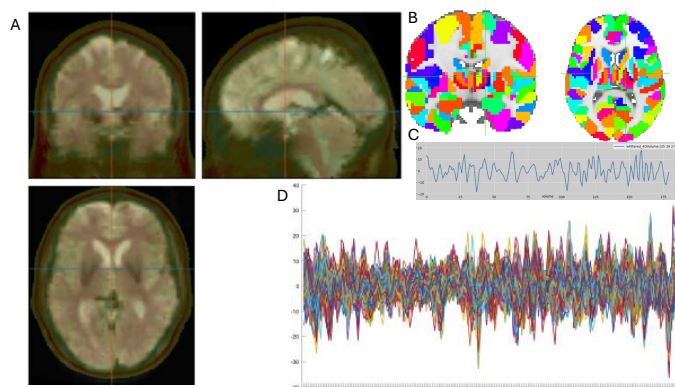


Figure 1. Example from a random subject 0211821. A: Normalization QC. B: Regions of Interest (ROIs) from the Brainnetome. C: Bold time series of the cursor position in B. D: Extracted time series from all 246 Brainnetome atlas ROIs.

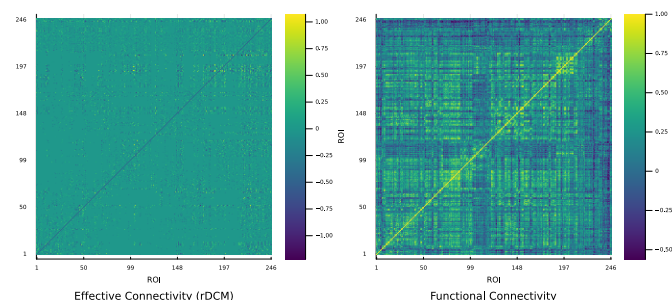


Figure 2. Functional and effective connectivity matrices for a single subject. Heatmaps depict pairwise connections among Brainnetome-defined Regions of Interest (ROI).

3.4.2. Regression Dynamic Causal Models

Regression Dynamic Causal Modeling (rDCM) was employed to investigate directed brain connectivity and estimate causal influences between brain regions at scale; rDCM represents a reformulation of the traditional DCM framework into a linear regression problem in the frequency domain, enabling efficient estimation even in large-scale brain networks [18]. The Brainnetome Atlas binary structural connectivity matrix across 246 regions was used to define the network architecture; this matrix served as a prior to constrain the model by indicating which region pairs are anatomically connected. For improved stability and convergence in rDCM estimation, the matrix was modified by setting diagonal elements (self-connections) to one. The strength and direction of connections among brain regions are estimated by fitting a generative model that explains observed BOLD signals through latent neural interactions (Fig. 2), allowing causal patterns in high-dimensional networks—potentially related to TBI and PTSD-related dysfunctions—to be explored.

3.5. Generative Embedding

After connectivity was estimated using both FC and rDCM, a generative embedding (GE) approach was applied to determine whether the resulting connectivity patterns could reveal group-level differences among Veterans with TBI, PTSD, and healthy controls. The inferred parameters from the generative models, connectivity estimates from rDCM or FC, were used as input features for downstream analyses. An initial dataset of 88 subjects comprised 26 with TBI, 20 with PTSD, 9 with comorbid TBI and PTSD, and 33 NC. For the first research question concerning the clustering, subjects with both TBI and PTSD were excluded from further analysis. To ensure balanced group sizes, 20 subjects were randomly selected from each of the three remaining groups (TBI, PTSD, and NC), resulting in a final dataset of 60 subjects.

3.5.1. Graph-Theoretic Metrics

A selection of graph-theoretic metrics was used to examine brain network changes potentially linked to cognitive decline. These metrics

were chosen based on the type of connectivity analyzed—functional (undirected) or effective (directed). For functional connectivity, metrics such as clustering coefficient, global efficiency, and betweenness centrality were used to assess network organization and communication efficiency. For effective connectivity, direction-sensitive measures like in- and out-strength, directed efficiency, and PageRank centrality were applied to capture the flow of information across brain regions. This approach enabled the evaluation of both localized and distributed network dynamics relevant to understanding disruptions in cognitive function.

3.5.2. Dimensionality Reduction

The FC and rDCM estimates were vectorized for each subject, after which dimensionality reduction methods—including Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), and variational autoencoders (VAEs)—were applied prior to clustering to reduce feature dimensionality and facilitate the discovery of meaningful group structures.

A variational autoencoder (VAE) was implemented using the Flux deep learning library. The connectivity data from all 88 subjects were split into training and validation sets, and the VAE was trained exclusively on the training set. The encoder was composed of two fully connected layers with ReLU activation, followed by two parallel linear layers to estimate the mean and log-variance of the latent space. The decoder mirrored the encoder's architecture, employing a Bernoulli output distribution and a sigmoid activation function at the final layer. A latent space of dimension d was adopted, and the model was trained using the Adam optimizer (learning rate of $1e-3$) for 50 epochs with a batch size of 10. The variational lower bound loss—incorporating both reconstruction loss and Kullback–Leibler (KL) divergence—was minimized during training. Mean squared error (MSE), binary cross-entropy, and total loss were monitored on both training and validation sets to evaluate model performance and prevent overfitting.

3.5.3. Latent Space Interpretability Analysis

To evaluate the contribution of individual brain connections to the latent space structure, we traced back the first principal components (PCs) from the PCA of effective (rDCM) and functional connectivity estimates to the regional connections (Fig. 6). The first PCs, capturing the dominant variance in the connectivity matrices, were projected back onto the original connectivity space to identify connections with the highest influence. A data-driven thresholding approach was employed using Otsu's method to distinguish the most relevant connection weights. Connections exceeding this threshold were considered for further analysis as meaningful contributors to the latent structure.

To facilitate interpretation at a higher anatomical level, the 246 subregions defined by the brain atlas to 10 broader brain regions (e.g., thalamus, basal ganglia) were mapped using a predefined lookup table. For each pair of these broader regions, the number of supra-threshold connections were computed and normalized this count by the total number of possible connections between the regions. This normalization yielded a relative importance score, enabling the identification of region-to-region connectivity patterns that are most influential in shaping the latent representation and driving group clustering.

3.5.4. Clustering techniques

The following unsupervised clustering techniques were employed to identify natural groupings in the connectivity data: k-means, fuzzy C-means, and hierarchical Ward clustering. In k-means, each data point is assigned to a single cluster based on proximity to cluster centroids. In fuzzy C-means, partial membership in multiple clusters is allowed, providing a measure of uncertainty. In hierarchical Ward clustering, clusters are iteratively merged in a manner that minimizes within-cluster variance, producing a dendrogram of nested clusters. The number of clusters was fixed at three to correspond to the known

diagnostic groups. Clustering was performed without access to diagnostic labels, and post hoc alignment between clusters and true group labels was evaluated using the adjusted Rand index (ARI), normalized mutual information (NMI), and accuracy. Accuracy was determined by finding, via permutation of cluster assignments, the mapping that yielded the highest agreement with true diagnostic labels.

For each experimental configuration—defined by a specific clustering method, dimensionality reduction technique, and latent space dimensionality—50 independent iterations were conducted to assess statistical robustness. In each iteration, 20 subjects were randomly sampled from each diagnostic group (TBI, PTSD, and controls) to ensure balanced class representation. Dimensionality reduction was then applied to the sampled dataset, after which clustering was performed using the specified algorithm.

To establish a chance-level baseline, 100 additional iterations were performed in which cluster labels were randomly assigned to the selected subjects. The mean and standard deviation of accuracy across these random-label iterations were calculated to provide a reference for interpreting clustering performance relative to random guessing.

3.5.5. Regression vs. Classification: Rationale and Metric Selection

A key methodological decision was been the choice between a classification and a regression framework for predicting the cognitive decline via clinical scores on the 12-month follow-up visit. Initially, classification was considered as a potential approach, whereby subjects could be grouped into discrete clinical categories (e.g. normal cognition, mild cognitive impairment, and stages of dementia). From this point, based on the jump of a patient from one category to another, a subject can be categorized as "decliner" a "non-decliner" based on the difference of the metric at the baseline visit and the predicted value for the follow-up. However, this strategy was ultimately deemed suboptimal for several reasons. First, cognitive outcomes such as MMSE scores are fundamentally continuous measures, and discretizing them can result in substantial information loss and reduced sensitivity to subtle changes. Second, since our dataset is highly unbalanced with respect to possible clinical outcome classes, it is non-trivial to train robust and generalizable classifiers. Imbalanced classes would exacerbate issues such as inflated accuracy due to majority class prediction, and complicate both model evaluation and interpretation.

To mitigate the aforementioned constraints, using regression models would be more appropriate, enabling, thus, direct modeling of continuous cognitive outcomes. In implementing regression models, a further decision concerned the choice of the evaluation metric for the model selection. While the root mean squared error (RMSE) is a standard choice for quantifying model fit in regression, given the difficulty of our prediction, it has been observed in the early stages of the model development that optimizing for the RMSE, very often gives negative coefficient of determination (R^2), which instantly rings a bell that something is not right.

(R^2) measures the proportion of variance in the observed outcome y explained by a linear regression model. Given n observations $\{(x_i, y_i)\}_{i=1}^n$ and fitted values $\hat{y}_i$, define:

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2, \quad SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

where $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$. The R^2 statistic is then

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SST - SSE}{SST}.$$

Optimization was therefore conducted using R^2 instead. A standardized measure of the variance in the outcome explained by the model, R^2 facilitates comparisons across models and datasets. Moreover, benchmarking against a null model (i.e., one that always predicts the mean outcome based on training-set statistics) is made straightforward: $R^2 = 0$ indicates no improvement over the null model, $R^2 = 1$

indicates perfect prediction, and negative values signify performance worse than the null model, thus yielding a transparent and clinically meaningful assessment of predictive utility.

3.5.6. Choice of Machine Learning Models

Given the limited sample size in this study (88 subjects, with 70 for training and 18 for testing, held-out), models were selected with an emphasis on interpretability, resistance to overfitting, and suitability for different data dimensionalities. Elastic net regression was used as the primary linear approach, because it combines L1 (lasso) and L2 (ridge) regularization. This method is particularly useful for datasets where many features may be correlated or irrelevant, as it can select important predictors while maintaining stable estimates.

To test whether more flexible models could improve predictive accuracy, two additional methods were applied. Random forest regression was chosen because it can automatically capture complex, nonlinear relationships between features and the outcome, and it is less prone to overfitting than many other nonlinear models, especially when the dataset is not large. Support vector regression (SVR) was also implemented, using linear, polynomial, and radial basis function kernels, to examine whether the outcome could be better predicted by more complex relationships between features.

The choice of these models reflects the need for data efficient model and the characteristics of the input data: connectivity matrices contain many variables and may encode subtle, distributed patterns relevant to cognitive decline, while the clinical and psychometric variables are fewer and more straightforward. By applying both linear and nonlinear models, the analysis was designed to balance interpretability with the potential to discover meaningful patterns that might not be captured by simpler methods. Neural networks were not considered, as their performance is strongly limited by small sample sizes and they are highly prone to overfitting in datasets of this scale.

3.5.7. Prediction approach - Mixture of Experts strategy

A stacked, multi-expert modeling framework was chosen to leverage complementary information from four distinct feature domains (questionnaire data, functional connectivity (FC), effective connectivity (rDCM), and APOE genotype) while minimizing overfitting in the context of a relatively small sample size. For each feature domain ("expert"), five base-learner algorithms were trained independently: (1) a linear regression model with elastic net regularization, (2) three support vector machines of different kernels (RBF, linear, polynomial), and (3) a random forest ensemble. Hyperparameter tuning for each base learner was performed via an inner 3-fold cross-validation loop, and performance was assessed in an outer 5-fold cross-validation loop. This nested CV structure ensured that hyperparameter selection remained strictly within training data on each fold. Concretely, within each outer fold (5-fold CV), the training subset was further partitioned into three folds. For each base learner on a given expert's features, optimal hyperparameters (e.g., elastic net mixing and penalty parameters, SVM cost and kernel width, number of trees and maximum depth for random forests) were chosen to minimize cross-validated error within that inner loop. The base learners were then retrained on the full training subset using these tuned hyperparameters. This nested CV structure ensured that hyperparameter selection was performed exclusively on the training subset of each outer fold, thus preventing information leakage.

The mathematical description for the mixture of experts, in our case, is straightforward. Suppose that M experts ($M = 3$ for CQ2 and 4 for CQ3) are trained, with each expert using a different feature set (e.g. Questionnaires, FC, EC, or APOE). Given a new subject with features $\mathbf{x}$, the final prediction $\hat{y}$ is given by:

$$\hat{y} = \sum_{m=1}^M \pi_m(\mathbf{x}) \cdot f_m(\mathbf{x}_m)$$

where:

- $f_m(\mathbf{x}_m)$ is the prediction from expert m (trained on features $\mathbf{x}_m$, a subset or projection of $\mathbf{x}$),
- $\pi_m(\mathbf{x})$ is the (possibly learned) gating (weighting) function that assigns a weight to each expert's prediction, subject to $\sum_{m=1}^M \pi_m(\mathbf{x}) = 1$.

Based on the above, a second-stage “meta-learner” can be trained to combine the best experts with a positive R^2 out of each data modality into a final outcome prediction. The meta-learner can take the form of another model, usually a linear regressor. In the most trivial case, it can be average of the contributions, provided that we have more than one expert. This “meta-learner” at this stage can enforce sparsity among the expert inputs, allowing uninformative algorithm-modality pairings to be downweighted or excluded entirely and thereby facilitating model interpretability. Once the optimal hyperparameters for the meta-learner are chosen, it can be retrained on the entire outer-fold training set and then used to generate final predictions on the held-out test set of that fold.

3.5.8. Dimensionality reduction

For the connectivity data, dimensionality reduction was an essential preprocessing step. Even after applying masking to exclude redundant or unnecessary elements—such as self-connections and structurally irrelevant zeros, the number of features derived from each subject's connectivity matrix remained extremely high, exceeding 22,000. The downstream models will use the data after the dimensionality reduction using the method that will be found most appropriate when addressing CQ2.

3.5.9. Nested CV and assurance of no data leakage

To ensure rigorous model evaluation and eliminate any risk of data leakage, nested cross-validation for all models was implemented. The datasets were first split into distinct training and held-out test sets, and the held-out test set was reserved strictly for the final evaluation and was never accessed during model development. Within the training set, we employed nested k-fold cross-validation, where the data were partitioned into an outer loop (to simulate held-out validation folds) and an inner loop (for hyperparameter tuning and model selection). This two-level approach guarantees that each outer validation fold is never used for model fitting or tuning.

Crucially, all data preprocessing steps, especially feature normalization, were performed within each fold of the cross-validation, using only the data in the current training subset of that fold. This means that the mean and standard deviation used to normalize features were computed solely from the training data of each fold and were then applied to the validation data. No information from the outer validation fold or the ultimate test set was used in preprocessing at any stage. If normalization were performed on the entire dataset prior to cross-validation, information from the test sets would inadvertently be introduced into the training process, resulting in optimistically biased performance estimates. By adhering to this strict separation, our approach ensures that the performance metrics reported represent a true, unbiased estimate of out-of-sample generalization performance. This methodology is considered best practice in machine learning and is especially critical for biomedical datasets, where overfitting is a substantial risk. Alternative approaches, such as simple cross-validation or bootstrapping without a held-out test set, do not provide the same safeguards against data leakage. Our strategy of nested k-fold cross-validation, combined with proper fold-wise normalization and a truly held-out test set, thus represents the most robust and defensible validation pipeline for our study.

3.5.10. Bootstrapping

Given the small size of our test set ($n = 18$), bootstrap resampling was used to estimate the confidence intervals for performance metrics such as RMSE, MAE, and R^2 on the final prediction. The bootstrap approach provides a non-parametric way to assess the uncertainty and variability of these metrics, even when the sample size is limited

and standard parametric assumptions may not hold. In this context, reporting p-values for model performance would not provide meaningful additional insight, as such values are unstable and unreliable with small samples. Instead, bootstrap confidence intervals offer a more informative and appropriate assessment of the uncertainty in our results. However, it should be emphasized that with such a small test set, the resulting confidence intervals are inherently wide and reflect the high sampling variability present in the data.

3.5.11. Interpretability Analysis

To further traceback the clustering information on the regional level in Q1, the first components of the PCA were mapped back on the incoming and outgoing brain regions from the brainnetome atlas using the circularGraph v2.0 package in matlab besides the plotted maps for FC and rDCM. Similar interpretability analyses for t-SNE and VAE remain to be explored in future work. Regarding Q2 and Q3, after training and validation of the MoE, interpretability methods were chosen based on the nature of the final meta-learner. If a linear Elastic Net model proved optimal, global interpretability was obtained by directly examining its fitted coefficients, since each coefficient's magnitude and sign transparently indicate that feature's contribution to the prediction. Conversely, if a non-linear model (SVM or Random Forest) emerged as the best model, then local, case-specific explanations were generated using LIME [16] or SHAP [19]. In that scenario, LIME or SHAP approximates the model's decision surface around individual test instances to attribute importance scores to input features. By selecting the interpretability technique conditionally, direct coefficient inspection for linear models, LIME or SHAP for complex models, it is ensured that the explanations matches the meta-learner's complexity and that the role of each feature or feature set in driving predictions is made transparent.

4. Results

4.1. Dataset

After the application of the inclusion criteria, 88 subjects remained. Demographic and clinical variables are stated for each group. No significant differences emerged between groups on any of the assessed variables (see Table 2). As shown in Figure 3, the distributions of MMSE, CDR-SOB, and GDS-15 scores are depicted for baseline (green) and 12-month follow-up (red), with black vertical lines indicating established clinically meaningful cut-offs.

	NC	PTSD	TBI	PTSD+TBI	p-value
<i>n</i>	33	20	26	9	—
<i>Age</i>	70.52 (5.98)	68.94 (2.31)	69.96 (4.15)	68.56 (4.14)	0.56
<i>CDR-SOB</i>	0.52 (0.97)	0.31 (0.46)	0.78 (0.82)	0.28 (0.57)	0.34
<i>GDS 15</i>	2.06 (2.19)	3.35 (2.76)	2.69 (2.62)	4.00 (2.74)	0.289
<i>MMSE</i>	28.4 (1.52)	28.2 (1.50)	27.9 (2.26)	27.2 (2.17)	0.472
<i>CAPS</i>	25.5 (25.7)	54.2 (12.6)	25.3 (22.1)	37.1 (16.2)	0.381

Table 2. Summary statistics by patient group (values are mean $\pm$ SD; n = count), p-values indicate significance of group differences in one-way ANOVA.

4.2. Q1: Clustering

Figure 4 illustrates how both the choice of dimensionality reduction method and the size of the reduced feature space influence clustering performance. For both rDCM and FC-derived connectivity data, clustering accuracy consistently exceeds chance-level performance, indicating that the connectivity features carry meaningful group-level information. This is supported by higher accuracy scores compared to the random baseline ($p < 0.05$). Notably, without applying dimensionality reduction—when using all pairwise connectivity weights directly—clustering performance remains significantly above chance ($p < 0.05$). Similarly, using graph-theoretic metrics alone also results in accuracy levels that surpass the random baseline, confirming the discriminative value of these features in distinguishing between clinical groups.

No statistically significant differences in clustering performance were found across dimensionality reduction methods (PCA, t-SNE,

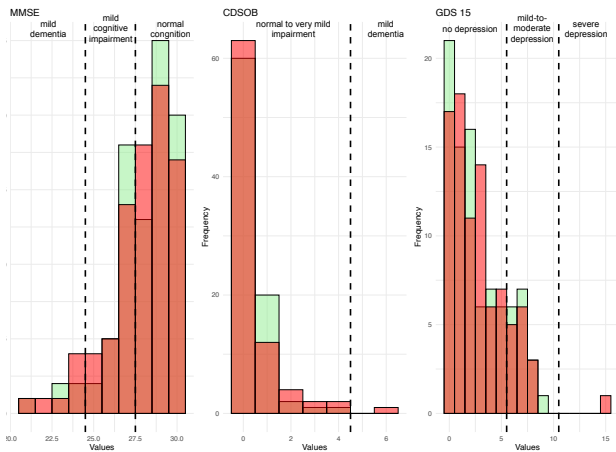


Figure 3. Outcome Measures of Cohort. Green at baseline, Red at 12 months follow-up, Black vertical lines indicate established clinical meaningful cut-offs as stated in the description of the clinical outcome measures

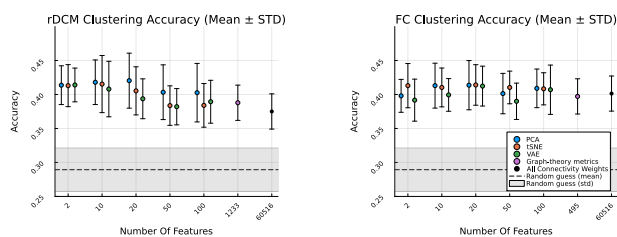


Figure 4. Comparison of the effects of dimensionality reduction techniques (PCA, t-SNE, and VAE) and latent space dimensionality on clustering accuracy.

VAE) or latent dimensions (2-100), as assessed by Kruskal-Wallis tests (all $p > 0.4$). Moreover, comparisons with the original data (without dimensionality reduction) also showed no significant differences. These results suggest that clustering performance remains robust regardless of dimensionality reduction strategy or the choice to reduce dimensions at all, within the tested parameters. When comparing clustering performance between rDCM and FC data across different settings, no statistically significant differences were found (all p -values ≥ 0.5) both in individual method/dimension comparisons and in the aggregate analysis.

Figure 5 shows that, in most cases, there is no statistically significant difference between the clustering techniques ($p > 0.1$). An exception is fuzzy k-means applied to PCA on the FC data, which performs notably worse. Nonetheless, all clustering methods consistently outperform random guessing. The first components for FC and rDCM are visualized in Figure 6.

4.3. Q2 and Q3: Mixed Expert Prediction Models

The overarching aim of this part of the study is to predict cognitive decline at 12-month follow-up, as measured by the MMSE, GDS and CDR-SOB scores, in our cohort of subjects with PTSD and/or TBI and controls. To achieve this, we leveraged three complementary sources

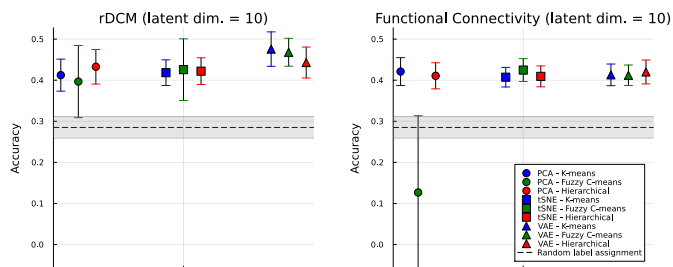


Figure 5. Comparison of the effects of clustering algorithms (K-means, Fuzzy C-means and Hierarchical Clustering) and dimensionality reduction techniques (PCA, t-SNE and VAE) on clustering accuracy

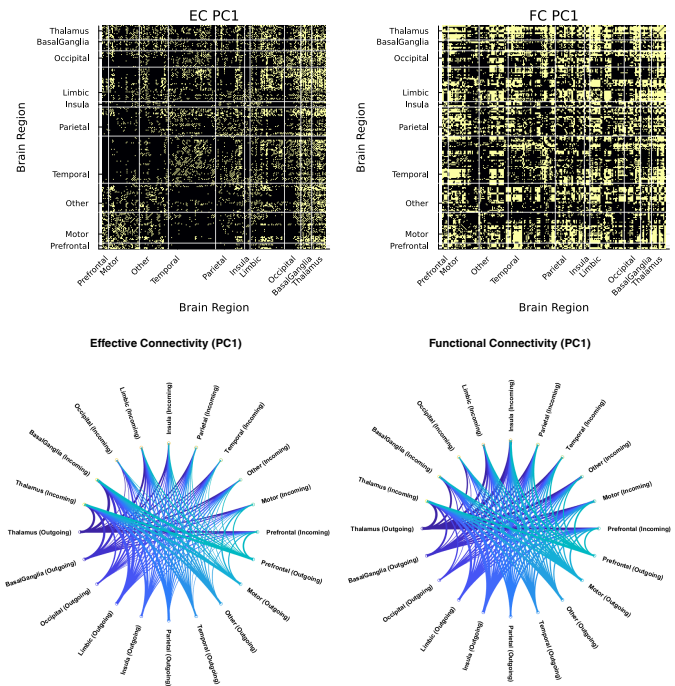


Figure 6. Regional trace back of the first Principal Components from the rDCM effective connectivity and functional connectivity estimates.

of baseline data: (1) clinical and demographic features extracted from standardized medical and neuropsychological questionnaires; (2) functional connectivity (FC) matrices derived from the resting-state fMRI imaging data (representing undirected correlations between brain regions); (3) effective connectivity (EC) matrices derived via the rDCM analysis, reflecting directed network influences; and (4) genetic risk factors, regarding the APOE status. Our key hypothesis was that advanced machine learning approaches, trained on all the aforementioned modalities, could identify meaningful patterns predictive of future cognitive trajectories.

4.3.1. MMSE

As shown in Table 3, the R^2 results after training via 5-fold nested cross-validation for each model and feature domain are reported. A positive R^2 indicates variance explained above the mean-only baseline, whereas negative values denote performance worse than predicting the group mean.

Model	Questionnaires	FC	rDCM	APOE
ENET	0.0821	-0.1100	-0.1661	-0.1004
SVM RBF	-0.0128	-0.0514	-0.1016	-0.1492
SVM Poly	-0.2856	-0.9588	-0.4124	-0.1749
SVM Linear	-0.1080	-0.1146	-0.1178	-0.1068
RFT	-0.2790	-0.2372	-0.1344	-0.0656

Table 3. R^2 values of different models across feature domains (Questionnaires, FC, rDCM, APOE) for predicting MMSE. ENET = Elastic Net regularization; SVM RBF = Support Vector Machine with radial basis function kernel; SVM Poly = SVM with polynomial kernel; SVM Linear = SVM with linear kernel; RFT = Random Forest Trees. Feature domains include Questionnaires, FC = functional connectivity, rDCM = regression Dynamic Causal Modeling, and APOE = apolipoprotein E genotype.

The best expert, the elastic net regularized regression model trained purely on the questionnaires, achieved a mean R^2 of 0.0821 (per-fold scores: -0.0413, 0.1460, -0.1799, 0.0906, 0.3952), indicating a small but positive ability to explain variance above the null model. Its average prediction error was quantified by an RMSE of 1.7481 (per-fold: 1.4741, 1.7712, 2.6562, 1.5048, 1.3344) and an MAE of 1.3287 (per-fold: 1.2255, 1.3927, 1.8343, 1.1572, 1.0338). Across folds, the most commonly selected hyperparameters were $\alpha=1.00$ (pure LASSO penalty) and $\lambda=0.37276$, demonstrating a preference

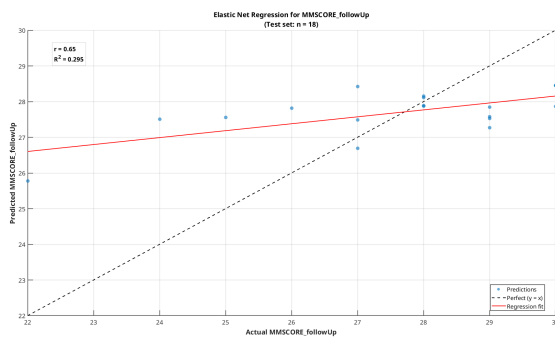


Figure 7. MMSE prediction on the test set.

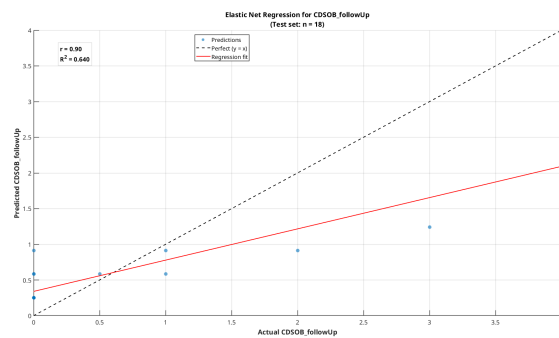


Figure 8. CDR-SOB prediction on the test set.

for sparse feature selection.

The Elastic Net model trained exclusively on questionnaire data (with $\alpha = 1.00$ and $\lambda = 0.37276$) was retrained on the full training set and evaluated on the held-out test set. On this test set, the model achieved an RMSE of 1.775 (bootstrap 95% CI: 1.221–2.272), an MAE of 1.411 (bootstrap 95% CI: 0.923–1.926), and an R^2 of 0.295 (bootstrap 95% CI: –0.203–0.442) (see Figure 7). By comparison, the null model (predicting the training-set mean) yielded an R^2 of –0.029 on the same test set. This indicates that the Elastic Net questionnaire expert captured approximately 29.5% of the variance in follow-up MMSE scores and substantially outperformed the null baseline.

As the best expert is the linear regression model with elastic net regularization, interpretability is most transparently achieved by examining the model coefficients themselves, and they directly give the contribution of each feature's effect on the prediction. In this context, the application of LIME, SHAP would not provide additional explanatory value, as its local surrogate models would simply reproduce these global coefficients for every test case, offering no further insight over the direct examination of the coefficients. For this reason, the interpretability analysis was ultimately focused on the global coefficients, which comprehensively summarize feature importance and the model's predictive behaviour. The final MoE elastic net predicting for the 12 months follow-up MMSE retained two nonzero features, MMSCORE_baseline with coefficient 0.4920 and ANARTERR with coefficient –0.0243. ANARTERR denotes the number of errors on the American National Adult Reading Test, serving as an index of premorbid verbal intelligence and where higher error counts predict slightly lower follow-up MMSE scores.

4.3.2. CDR-SOB

As shown in Table 4, the R^2 results after training via 5-fold nested cross-validation for each model and feature domain are reported.

Model	Questionnaires	FC	rDCM	APOE
ENET	0.3178	-0.0274	-0.0766	-1.4135
SVM RBF	0.0663	-0.0857	-0.6042	-0.0257
SVM Poly	-3.7801	-1.0160	-2.2543	-0.1594
SVM Linear	-2.9903	-0.1114	-0.4342	-0.2187
RFT	0.1676	-0.4319	-0.3099	-0.2546

Table 4. R^2 values of different models across feature domains (Questionnaires, FC, rDCM, APOE) for predicting CDR-SOB. ENET = Elastic Net regularization; SVM RBF = Support Vector Machine with radial basis function kernel; SVM Poly = SVM with polynomial kernel; SVM Linear = SVM with linear kernel; RFT = Random Forest Trees. Feature domains include Questionnaires, FC = functional connectivity, rDCM = regression Dynamic Causal Modeling, and APOE = apolipoprotein E genotype.

The best Elastic Net expert achieved a mean R^2 of 0.3178 (per-fold scores: 0.2722, 0.2889, 0.1596, 0.0026, 0.8657), indicating a substantial ability to explain variance above the null model. Its average prediction error was quantified by an RMSE of 0.7462 (per-fold: 1.4369, 0.8064, 0.6589, 0.4098, 0.4191) and an MAE of 0.5088 (per-fold: 0.7265, 0.5982, 0.5192, 0.3722, 0.3280). Across folds, the most commonly selected

hyperparameters were $\alpha = 0.70$ and $\lambda = 0.37276$, reflecting a balance between sparsity and coefficient shrinkage.

The Elastic Net model trained on CDR-SOB data (with $\alpha = 0.70$ and $\lambda = 0.37276$) was retrained on the full training set and evaluated on the held-out test set. On this test set, the model achieved an RMSE of 0.692 (bootstrap 95% CI: 0.396–0.951), an MAE of 0.513 (bootstrap 95% CI: 0.326–0.746), and an R^2 of 0.640 (bootstrap 95% CI: –0.116–0.776) (see Figure 8). By comparison, the null model (predicting the training-set mean) yielded an R^2 of –0.002 on the same test set. This indicates that the Elastic Net CDR-SOB expert captured approximately 64.0% of the variance in follow-up CDR-SOB scores and significantly outperformed the null baseline.

The Elastic Net model predicting follow-up CDR-SOB retained a single nonzero feature, CDR-SOB at baseline with coefficient 0.5119. Baseline CDR-SOB quantifies initial dementia severity (sum of ratings across memory, orientation, judgment, community affairs, home and hobbies, and personal care), and higher baseline scores predict proportionally higher CDR-SOB at follow-up.

4.3.3. GDS

As shown in Tables 5, the R^2 results after training via 5-fold nested cross-validation for each model and feature domain are reported.

Model	Questionnaires	FC	rDCM	APOE
ENET	0.3582	-0.0370	-0.0759	-0.2739
SVM RBF	0.2749	-0.3233	-0.1088	-0.0832
SVM Poly	-0.2933	-4.2565	-0.0092	-0.1036
SVM Linear	0.3362	-0.1219	-0.3873	-0.3761
RFT	0.2411	-0.3735	-0.1179	-0.1079

Table 5. R^2 values of different models across feature domains (Questionnaires, FC, rDCM, APOE) for predicting GDS. ENET = Elastic Net regularization; SVM RBF = Support Vector Machine with radial basis function kernel; SVM Poly = SVM with polynomial kernel; SVM Linear = SVM with linear kernel; RFT = Random Forest Trees. Feature domains include Questionnaires, FC = functional connectivity, rDCM = regression Dynamic Causal Modeling, and APOE = apolipoprotein E genotype.

The best Elastic Net expert achieved a mean R^2 of 0.3582 (per-fold scores: 0.1486, 0.5280, 0.3813, 0.1836, 0.5496), indicating a strong ability to explain variance above the null model. Its average prediction error was quantified by an RMSE of 2.1388 (per-fold: 3.4806, 1.8518, 1.8911, 1.7282, 1.7424) and an MAE of 1.5591 (per-fold: 1.9198, 1.5444, 1.5698, 1.3916, 1.3700). Across folds, the most commonly selected hyperparameters were $\alpha = 0.90$ and $\lambda = 0.37276$, reflecting a balance between sparsity and coefficient shrinkage.

The Elastic Net model trained on GDS-15 data (with $\alpha = 0.90$ and $\lambda = 0.37276$) was retrained on the full training set and evaluated on the held-out test set. On this test set, the model achieved an RMSE of 1.572 (bootstrap 95% CI: 1.041–2.077), an MAE of 1.259 (bootstrap 95% CI: 0.846–1.742), and an R^2 of 0.516 (bootstrap 95% CI: –0.104–0.762) (see Figure 9). By comparison, the null model (predicting the training-set mean) yielded an R^2 of –0.012 on the same test set. This indicates that the Elastic Net GDS-15 expert captured

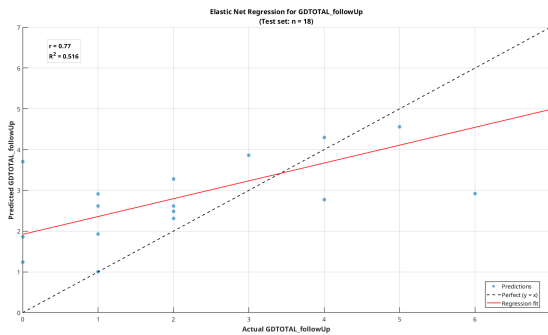


Figure 9. GDS 15 prediction on the test set.

approximately 51.6% of the variance in follow-up GDS scores and substantially outperformed the null baseline.

The Elastic Net model predicting follow-up GDS retained three nonzero features, GDS at baseline with coefficient 1.1587, PATIENT-GROUP with coefficient -0.1691 , CAPSSCORELIFE with coefficient 0.2562, and CAPSSCORECURR with coefficient 0.0979. GDTOTAL is the total score on the Geriatric Depression Scale at baseline, while CAPSSCORELIFE and CAPSSCORECURR are the lifetime and current scores, respectively, on the Clinician-Administered PTSD Scale; higher values on each feature predict higher follow-up GDS scores.

5. Discussion

5.1. Dataset

The ADNI DOD veteran cohort represents a rare and invaluable resource for understanding long-term neural and cognitive sequelae of TBI and PTSD, as it follows Vietnam War veterans aged 50 to 90, several decades after their initial injuries or combat exposure. By combining high-resolution structural MRI, resting-state fMRI, APOE genotyping, and detailed neuropsychological assessments with a 12-month follow-up, the dataset enables longitudinal prediction of late-life outcomes in a chronically exposed population, an opportunity seldom available in military cohorts. A notable methodological innovation in our analysis is the use of an atlas-based parcellation, namely the 246-region Brainnetome Atlas, rather than the more common spatial ICA approach, because whereas ICA derives data-driven networks whose spatial maps can vary across subjects, the fixed Brainnetome parcels facilitate direct between-subject comparisons and support computation of graph-theoretic metrics on a well-established structural scaffold. Moreover, the same atlas serves as the anatomical prior for regression DCM, allowing estimation of directed connectivity patterns constrained by known white-matter pathways, whereas ICA-derived time series lack a straightforward structural adjacency matrix, making them less amenable to large-scale DCM analyses, so this atlas approach not only improves reproducibility but also broadens the scope of network models to include effective connectivity, a domain that remains underexplored in PTSD and TBI research. Finally, the breadth of data modalities, including self-report and clinician-administered scales, resting-state connectivity (both functional and directed), and genotype, permits a comprehensive mixed-expert modeling framework, as few prior studies have integrated these diverse features into a single nested cross-validation pipeline to predict future cognitive or affective decline, and because many established risk factors, for example APOE $\epsilon 4$, exert effects over decades, having both a long post-injury interval and multiple baseline modalities strengthens our ability to identify which signatures truly forecast decline over a one-year horizon, so the ADNI DOD dataset's unique population, atlas-driven connectivity approach, and multimodal design together create novel opportunities to uncover the network-based and genetic determinants of late-life neuropsychiatric outcomes in veterans.

5.2. Q1: Clustering

The unsupervised clustering analyses demonstrated that baseline resting-state fMRI (rs-fMRI) connectivity patterns contain robust

group-level information capable of distinguishing Vietnam War veterans with TBI, PTSD, or healthy control (NC) status. Across multiple clustering algorithms (k-means, fuzzy C-means, and hierarchical Ward) and dimensionality reduction techniques (PCA, t-SNE, VAE), classification accuracy consistently exceeded the chance-level as a baseline, confirming that intrinsic network topology differentiates diagnostic groups. Similarly, graph-theoretic metrics (e.g., node strength, clustering coefficient, local/global efficiency) derived from functional or directed (rDCM) connectivity also yielded clustering performance surpassing the random baseline. These findings support the hypothesis that large-scale network changes are identifiable via rs-fMRI in veterans with TBI and PTSD. In particular, the absence of any significant difference in clustering accuracy across dimensionality reduction methods or latent dimensions suggests that discriminative information is distributed widely across connectivity edges rather than concentrated in a small subset of features. Likewise, no statistically significant difference was observed between clustering on FC versus rDCM data, implying that both undirected correlation measures and directed (causal) influence metrics capture similar group-specific network alterations.

The resilience of clustering performance, despite modality (FC vs. rDCM), algorithmic choice, and dimensionality reduction strategy, underscores the presence of strong, biologically meaningful signatures of TBI and PTSD in rs-fMRI. These signatures likely reflect chronic adaptations of large-scale systems (e.g., default mode, salience, frontoparietal networks) to traumatic changes or stress. Prior literature has highlighted widespread structural white-matter and functional connectivity alterations following TBI (e.g. regulation of activity in the default mode network [7], [8]) and in chronic PTSD (e.g., increased amygdala–insula connectivity, decreased amygdala–hippocampus connectivity, and weakened amygdala–dorsal/rostral ACC anticorrelation [10], aberrant default mode network [4], [5] and central executive, salience network changes [12], [13], [15]). The present clustering results extend these observations by demonstrating that, even without supervised labels, group membership can be reliably inferred from connectivity alone. Thus, first PCA component captures the single strongest axis of variance in large-scale network organization, and its preserved clustering power confirms that group-specific network disruptions (e.g., subcortical–cortical dysregulation) are the primary driver of diagnostic differences (see Figure 6). In the effective connectivity (rDCM PC1) Prefrontal, Motor, and Insular parcels register relatively weaker weights, with dominated subcortical–cortical weights (Basal Ganglia, Thalamus and Temporal, Parietal, Others). Likewise, the FC PC1 shows that widespread thalamo–cortical and basal ganglia interactions account for most of the covariance in undirected correlations, with reduced loading in Prefrontal, Motor and Temporal regions. This is in line with potential disruptions in white-matter bundles and higher network associations. From a methodological standpoint, the lack of superiority for any particular dimensionality reduction or clustering method suggests that when group-level differences are strong, as in this chronic TBI or PTSD cohort, complex preprocessing steps (e.g., nonlinear embedding or autoencoder-based reduction) may not be necessary to reveal diagnostic structure. Furthermore, the fact that clustering on graph-theoretic summaries (rather than all pairwise edges) also produced above-chance accuracy indicates that higher-order topological features capture much of the discriminatory signal. In practical terms, this implies that future studies could reduce computational burden and improve interpretability by focusing on a parsimonious set of network metrics. Which has to be further investigated. Clinically, these clustering results validate the notion that intrinsic connectivity signatures remain altered even many years post-injury or post-trauma (veterans aged 50–90 in the DOD ADNI cohort). Because the current analyses were performed at baseline of the study many years after the war exposure, the clustering accuracy suggests that network disruptions precede and may ultimately predict subsequent cognitive or

functional decline. In other words, the same connectivity patterns that allow unsupervised group separation may serve as early biomarkers for identifying individuals at high risk for future degeneration.

In summary, Q1's clustering analyses demonstrate that rs-fMRI carries a stable, differentiable signature of TBI- and PTSD-related network dysfunction. These signatures are robust to choice of clustering algorithm, feature-space dimensionality, and connectivity modality (FC vs. rDCM). Consequently, large-scale network disruptions can be detected in a purely data-driven manner, providing strong justification for incorporating connectivity-based features into downstream supervised prediction models (Q2), as well as for exploring interactions with genetic (APOE) or biospecimen moderators (Q3) in future work.

5.3. Q2 and Q3: Mixed Expert Prediction Model

The mixed-expert modeling framework combined four distinct feature domains, questionnaire data, functional connectivity (FC), effective connectivity (rDCM), and APOE genotype, to predict 12-month cognitive and clinical outcomes (MMSE, CDR-SOB, GDS) in veterans with TBI, PTSD, or neither. By training separate base learner (ENET, SVM-RBF, SVM-Poly, SVM-Linear, RFT) on each modality and then fusing their outputs via a second-stage meta-learner, the approach can exploit complementary signal strengths while accommodating differing scales, noise characteristics, and predictive utility inherent to each data source. Among the four modalities, questionnaire data consistently yielded the strongest predictive performance. For MMSE follow-up, the Elastic Net questionnaire expert achieved a modest $R^2 = 0.295$, RMSE = 1.775, MAE = 1.411 on the test set, substantially outperforming baseline connectivity or genetic models. Similarly, for CDR-SOB, the Elastic Net expert based on questionnaires attained a $R^2 = 0.640$, RMSE = 0.692, MAE = 0.513, indicating that clinician-administered scales capture a large fraction of variance in 12-month dementia severity. For GDS-15, the Elastic Net questionnaire expert achieved a $R^2 = 0.516$, RMSE = 1.572, MAE = 1.259, confirming that baseline depressive symptoms strongly forecast future GDS scores. By contrast, FC- and rDCM-based experts generally produced negative mean R^2 values across all outcomes, indicating performance worse than the null model. This suggests that, within the 12-month horizon and given the sample size, resting-state connectivity alone lacks sufficient predictive power to predict individual cognitive or affective trajectories more accurately than a mean predictor. Similarly, APOE genotype contributed little predictive value across all models, too, likely reflecting its modest effect on short-term change and the cross-sectional nature of the design. Hence, given the dominance of the questionnaire data, the mixed expert approach degenerated to one single expert, the elastic net on questionnaire data, trivially giving it a weight of one, instead of a coefficient that would have been otherwise determined by the meta-learner. Based on the results of the experts trained on the connectivity and the APOE feature, none of them was satisfactory, as they did not outperform the null model. Therefore, their expertise would not add any value to the final prediction.

The nonzero regression coefficients for each outcome are examined to reveal which baseline features drive follow-up MMSE, CDR-SOB, and GDS scores. Baseline MMSE is a strong predictor of 12-month MMSE because it captures an individual's current global cognitive status; a higher baseline score indicates greater cognitive reserve and resilience, making decline less likely over a one-year interval. ANARTERR, the error count on the American National Adult Reading Test, serves as an estimate of premorbid verbal intelligence; fewer reading errors reflect higher baseline cognitive ability. Since premorbid IQ moderates the trajectory of cognitive decline, ANARTERR's negative coefficient indicates that individuals with more reading errors (lower estimated premorbid verbal IQ) tend to have lower follow-up MMSE scores. Together, these features account for both current cognitive performance (MMSE at baseline) and lifelong intellectual baseline (ANARTERR), thereby capturing both state and trait determinants of future global cognition. Baseline CDR-SOB was

identified as the sole nonzero predictor of follow-up CDR-SOB, with coefficient 0.5119. This indicates that initial dementia severity, as captured by the summed scores across memory, orientation, judgment, community affairs, home and hobbies, and personal care, strongly informs future cognitive staging. By reflecting an individual's current impairment level, baseline CDR-SOB provides a stable anchor for predicting progression over the ensuing year. Baseline GDS-15 total score emerged as the strongest predictor of follow-up GDS-15, with coefficient 1.1587, indicating that initial depressive symptom burden largely determines future symptom levels. Lifetime PTSD severity (CAPSSCORELIFE, coefficient 0.2562) and current PTSD severity (CAPSSCORECURR, coefficient 0.0979) also contributed positively, reflecting the known comorbidity between posttraumatic stress and depression. By contrast, the negative coefficient for PATIENTGROUP (−0.1691) suggests that group membership (e.g., PTSD vs. NC) modulates the predicted depressive trajectory, potentially capturing unmeasured factors from the professional grouping on top of the CAPS scores. Together, these features integrate baseline mood, chronic PTSD exposure, and diagnostic category to forecast one-year depression severity.

The disparate performance across modalities underscores the necessity of a mixture-of-experts (MoE) architecture. Each feature set, questionnaire responses, FC parameters, rDCM parameters, and APOE status, operates on different scales and exhibits distinct noise profiles. Questionnaire data are low-dimensional, clinically validated summary scores (e.g., MMSE, GDS, CDR-SOB) with established psychometric properties, yielding strong within-sample correlations with follow-up outcomes. Functional connectivity (FC) comprises thousands of pairwise Pearson correlations among 246 Brainnetome regions, and Effective connectivity (rDCM) estimates, as well, creating a high-dimensional feature space that may require much larger cohorts or longer follow-up intervals to stabilize predictive relationships even after reasonable dimensionality reduction. APOE genotype is a ternary genetic risk factor with well-known associations to late-life Alzheimer's disease, but its incremental predictive power for one-year change in a middle-aged to elderly cohort can be limited. Because these domains yield potentially partially orthogonal information and differ in dimensionality, variance explained, and noise characteristics, no single model or data type suffices. The MoE framework allows each modality to "specialize" via its best base learner, then fuses their outputs. Consequently, the final ensemble can capitalize on strong questionnaire signals without being dragged down by noise and here now excluded connectivity or genetic features. Concerning the figures 7, 8, 9, the black dashed lines have a slope of 1, representing the ideal "perfect prediction" (i.e., every predicted score value exactly equals the actual score). In contrast, the red regression lines for each outcome score has a shallower slope (well below 1). This lower incline indicates that the model's predictions tend to be compressed toward the means, meaning low actual scores are overestimated and high actual scores are underestimated. In other words, the elastic-net models do not track the full range of variability in follow-up clinical scores, which is why its red line rises more gradually compared to the 45° black line.

The predominantly cross-sectional nature of all feature sets at baseline and restricted variance of the clinical outcome measures over time limits predictive horizons. Homozygotes APOE $\epsilon 4$ allele carriage is a well-validated risk factor for Alzheimer's disease, but its effect on short-term (12-month) change in MMSE, CDR-SOB, or GDS within a heterogeneous veteran cohort as assessed in this study is minimal. Negative R^2 values for all APOE-based models suggest that, in this timeframe, APOE status does not meaningfully differentiate individuals experiencing subclinical decline from those who remain stable. Likewise, FC and rDCM, while reflecting current network disruptions from TBI or PTSD, may not capture incipient neurodegenerative processes that unfold over multiple years. Thus, the predictive utility of established risk factors, genetic or imaging, for imminent

cognitive decline is constrained by study design. Longitudinal cohorts spanning several years or decades would likely reveal stronger APOE-related effects. Additionally, the whole study was conducted multiple years after the potentially life changing burden of the war exposure. Therefore, initial changes immediately after return cannot be recorded in this study, nor can long-term developments that go beyond the 12 months.

Overall, the results highlight that within a one-year follow-up clinical scales remain the most reliable predictors of near-term cognitive and affective outcomes. Resting-state connectivity metrics (FC, rDCM) and APOE genotype did not outperform mean-prediction baselines, suggesting that larger sample sizes, longer follow-up durations, additional biomarkers, or alternative imaging modalities (e.g., task fMRI, diffusion MRI) may be required to glean predictive signal from neural substrates. Prospective validation of the MoE ensembles in independent samples should be performed, as the present nested CV results serve primarily as a proof of concept in a mid-sized dataset.

6. Appendix

6.1. Mathematical Framework of Regression DCM (rDCM)

rDCM is grounded in a generative modeling approach that aims to infer effective connectivity—i.e., the directed influences between brain regions—by modeling the underlying neuronal dynamics that give rise to observed fMRI signals. It reformulates the classical DCM equations into a regression framework in the frequency domain, significantly reducing computational burden.

In its linear form, the neuronal activity $\mathbf{x}(t) \in \mathbb{R}^N$ for N brain regions evolves according to:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{C}\mathbf{u}(t)$$

where: $\mathbf{A}$ is endogenous connectivity matrix (effective connections between brain regions). $\mathbf{C}$ is the driving input matrix (how external stimuli or internal fluctuations affect nodes). $\mathbf{u}(t)$ is the input vector (stimulus or endogenous fluctuations).

Using the Fourier transform and properties of differentiation in the frequency domain:

$$\mathcal{F}\left[\frac{d\mathbf{x}(t)}{dt}\right] = i\omega\hat{\mathbf{x}}(\omega)$$

This yields the following algebraic relation:

$$i\omega\hat{\mathbf{x}}(\omega) = \mathbf{A}\hat{\mathbf{x}}(\omega) + \mathbf{C}\hat{\mathbf{u}}(\omega)$$

Rearranged, we get:

$$\hat{\mathbf{x}}(\omega) = (i\omega\mathbf{I} - \mathbf{A})^{-1}\mathbf{C}\hat{\mathbf{u}}(\omega)$$

This equation shows how neuronal responses in the frequency domain depend on the underlying effective connectivity structure and external inputs.

Rather than using a complex, nonlinear hemodynamic model as in classical DCM, rDCM assumes a linear convolution with a fixed hemodynamic response function (HRF):

$$\hat{\mathbf{y}}(\omega) = \hat{h}(\omega) \cdot \hat{\mathbf{x}}(\omega) + \mathbf{v}$$

where $\hat{h}(\omega)$ is the frequency-domain HRF and $\mathbf{v}$ is observation noise (assumed Gaussian with region-specific variances)

This linearized model allows the full system to be cast as a general linear model (GLM):

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \tau^{-1}\mathbf{I})$$

where $\mathbf{Y}$ is the observed BOLD signal in frequency domain, $\mathbf{X}$ is the design matrix encoding system dynamics (based on $\hat{\mathbf{x}}$ and inputs), $\boldsymbol{\theta}$ is the vector of model parameters (elements from $\mathbf{A}$ and $\mathbf{C}$) and τ is the noise precision (inverse variance).

Model parameters are estimated via Variational Bayes under a conjugate prior setup: $\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ and $\tau \sim \text{Gamma}(\alpha_0, \beta_0)$.

Posterior distributions for $\boldsymbol{\theta}$ and τ are then updated iteratively:

$$\boldsymbol{\Sigma}_{\boldsymbol{\theta}} = \left(\boldsymbol{\Sigma}_0^{-1} + \tau\mathbf{X}^T\mathbf{X}\right)^{-1}$$

$$\boldsymbol{\mu}_{\boldsymbol{\theta}} = \boldsymbol{\Sigma}_{\boldsymbol{\theta}} \left(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0 + \tau\mathbf{X}^T\mathbf{Y}\right)$$

Noise precision is updated via:

$$\alpha = \alpha_0 + \frac{N}{2}, \quad \beta = \beta_0 + \frac{1}{2} (\|\mathbf{Y} - \mathbf{X}\boldsymbol{\mu}_{\boldsymbol{\theta}}\|^2 + \text{tr}(\mathbf{X}^T\mathbf{X}\boldsymbol{\Sigma}_{\boldsymbol{\theta}}))$$

This iterative scheme continues until convergence and produces both parameter estimates and a lower bound on model evidence, allowing model comparison and selection.

6.2. Mathematical Framework for Baseline Functional Connectivity Estimation

Functional connectivity captures statistical dependencies between neural signals. When using Pearson correlation, FC between two regions i and j is quantified by:

$$W_{ij} = \frac{\text{cov}(Y_i, Y_j)}{\sigma_{Y_i}\sigma_{Y_j}}$$

where x_i and x_j are the preprocessed time series from regions of interest (ROIs) i and j , respectively, and σ denotes the standard deviation. The resulting correlation matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$, with N being the number of ROIs, is symmetric and bounded between -1 and 1 . To ensure numerical stability in downstream analysis, we apply Fisher z-transformation:

$$z_{ij} = \tanh^{-1}(r_{ij}) = \frac{1}{2} \ln \left(\frac{1 + r_{ij}}{1 - r_{ij}} \right)$$

This transformation normalizes the distribution of correlation coefficients, useful when comparing correlations across subjects.

6.3. Mathematical Formulation of Generative Embedding for Group Stratification

After computing subject-specific brain connectivity matrices using either FC or rDCM, we construct a feature embedding space to uncover latent group structure through unsupervised learning.

Let $\mathcal{D} = \{\mathbf{C}_s\}_{s=1}^S$ denote the set of connectivity matrices from S subjects, where each matrix $\mathbf{C}_s \in \mathbb{R}^{N \times N}$ encodes pairwise connections between N regions of interest (ROIs).

For FC, $\mathbf{C}_s$ is symmetric and contains Pearson correlation coefficients. For rDCM, $\mathbf{C}_s$ represents directed effective connectivity and is not necessarily symmetric.

Each matrix is vectorized to produce a subject-specific feature vector:

$$\mathbf{x}_s = \text{vec}(\mathbf{C}_s) \in \mathbb{R}^D, \quad D = N(N-1) \text{ (for rDCM)}, \quad D = \frac{N(N-1)}{2} \text{ (for FC)}$$

These vectors form the dataset:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_S]^T \in \mathbb{R}^{S \times D}$$

To uncover latent subgroups corresponding to TBI, PTSD, and control cohorts, we fit a GMM to $\mathbf{X}$. The GMM assumes that the data is generated from a mixture of K multivariate Gaussian components:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

where π_k is the mixing weight, satisfying $\sum_{k=1}^K \pi_k = 1$, $\boldsymbol{\mu}_k \in \mathbb{R}^D$ is the mean of cluster k , $\boldsymbol{\Sigma}_k \in \mathbb{R}^{D \times D}$ is the covariance of cluster k .

Cluster assignments $z_s \in \{1, \dots, K\}$ for each subject are determined by the posterior:

$$z_s = \arg \max_k \gamma_{s,k}, \quad \text{where } \gamma_{s,k} = \frac{\pi_k \mathcal{N}(\mathbf{x}_s | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_s | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

For visualization or regularization, we reduce dimensionality before clustering using PCA or t-SNE.

PCA: Compute the top $d \ll D$ principal components $\mathbf{U}_d \in \mathbb{R}^{D \times d}$ and project:

$$\mathbf{x}_s^{(PCA)} = \mathbf{U}_d^T (\mathbf{x}_s - \bar{\mathbf{x}})$$

t-SNE: Optimize a low-dimensional embedding that preserves local neighborhood similarity.

We assess alignment between the cluster assignments $\mathbf{z} = (z_1, \dots, z_S)$ and the ground truth labels $\mathbf{y} = (y_1, \dots, y_S)$ (TBI, PTSD, control) using Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI).

$$\text{ARI} = \frac{\text{Index} - \text{Expected Index}}{\text{Max Index} - \text{Expected Index}} \in [-1, 1]$$

$$\text{NMI}(\mathbf{z}, \mathbf{y}) = \frac{2I(\mathbf{z}; \mathbf{y})}{H(\mathbf{z}) + H(\mathbf{y})}$$

Higher ARI and NMI indicate better agreement between discovered clusters and known diagnostic groups.

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